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Beyond frequency: a capacity-weighted and reliability-adjusted framework for measuring effective public transport accessibility

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Abstract

The Public Transport Accessibility Level (PTAL) metric is gaining traction in Australian planning policy, most recently in Victoria's December 2025 reform of parking provisions, and represents a genuine advance in evidence-based land use decision-making. However, PTAL's reliance on service frequency and walking distance alone, treating all services as equivalent regardless of vehicle capacity or reliability, limits its ability to reflect real-world transport performance. In multi-modal networks like South East Queensland (SEQ), these limitations can materially misrepresent the people-moving capacity available to a location.

This paper introduces the Public Enhanced Transport Accessibility Level (PETAL): an evolution of PTAL that builds on its foundations by incorporating vehicle capacity, service reliability, route hierarchy, and bidirectional accessibility. Using open General Transit Feed Specification (GTFS) data and spatial analysis, PETAL measures effective accessibility as capacity per hour, operationalised against supportable bedroom density to produce a metric directly applicable to planning decisions.

Applied to SEQ, PETAL provides a more defensible basis for identifying land use mismatches: where zoning underdelivers relative to network capacity, where infrastructure investment is underleveraged, and where bedroom density supportable by the network diverges from what planning rules allow. The flood constraint analysis represents a novel contribution: identifying where high-accessibility land is effectively constrained by flood risk, with direct implications for infrastructure investment sequencing and climate adaptation planning. Better alignment between transit capacity and land use regulation directly supports housing supply and economic productivity.

The core methodological contribution - measuring effective accessibility as capacity per hour rather than services per hour - has direct application to housing policy, density controls, and network investment prioritisation. PETAL is offered as a transparent, scalable, and policy-ready evolution of existing PTAL practice, transferable to any Australian or international city with open GTFS data seeking a more rigorous basis for transit-informed land use policy.

Word count: 248

1. Introduction

The Public Transport Accessibility Level (PTAL) metric was originally developed as a simple, standardised way to quantify access to public transport based on walk distance and service frequency (Transport for London, 2015). Over time, it has become one of the most widely used accessibility indicators in planning practice, valued for its transparency, ease of application, and ability to translate complex transport networks into a single, policy-relevant measure. By reducing accessibility to a comparable index, PTAL has enabled planners to link land use outcomes, such as parking provision, density, and development intensity, to the availability of public transport in a consistent and defensible manner.

At its core, PTAL reflects a shift away from static, car-based planning assumptions toward a more dynamic understanding of accessibility grounded in public transport supply. It provides a practical bridge between transport modelling and statutory planning, allowing accessibility to be embedded directly into policy frameworks without requiring complex modelling inputs. This has made PTAL particularly attractive for jurisdictions seeking to operationalise transit-oriented development principles in a clear and implementable way.

The adoption of PTAL in Australian planning policy represents a genuine and welcome advance. In New South Wales, PTAL is already embedded within the Movement and Place Framework as a core built environment indicator used to assess the level of connectivity between land use and public transport services. It is used to support planning decisions, including identifying suitable locations for higher-density development, informing infrastructure investment, and guiding parking provision based on accessibility outcomes. Victoria's December 2025 reform of Clause 52.06 of the Victoria Planning Provisions, linking car parking requirements directly to measured transit accessibility, is the most recent and significant example of PTAL being embedded as a statutory planning instrument in Australia. Collectively, these developments signal a broader shift toward evidence-based land use policy that deserves recognition.

The economic case for getting this right is substantial. The Queensland Productivity Commission (2025, p. 235) estimated that targeted zoning reforms in well-located areas around transport networks in SEQ could deliver net benefits of up to \$48 billion, more than double the \$18 billion projected from dispersed growth. Realising those benefits requires first making the opportunities legible: identifying where transit capacity already exists, where density policy fails to match it, and where constraints such as flood risk limit what is achievable. PETAL Explorer is designed to do exactly that.

The question this paper addresses is not whether PTAL is useful—it clearly is—but whether service frequency alone captures what planners need to know. A location served by ten trains per hour is not equivalent to one served by ten buses per hour, nor is a reliable busway service comparable to an unreliable mixed-traffic route operating at the same scheduled frequency. Standard PTAL treats these scenarios as equivalent. In SEQ's multi-modal network, comprising heavy rail, busway, G:link light rail, CityCat ferries, and conventional bus services, this distinction is critical. It directly influences which locations are identified as suitable for higher-density development, which infrastructure gaps are visible to policy makers, and which planning decisions can be robustly justified under scrutiny.

This paper introduces the Public Enhanced Transport Accessibility Level (PETAL): an evolution of PTAL developed and applied in SEQ that incorporates vehicle capacity, service reliability, route hierarchy, and bidirectional accessibility. The methodology is implemented in PETAL Explorer (petalexplorer.org, 2026), an open-source, publicly accessible tool covering Brisbane and the Gold Coast at 100m grid resolution, with planning overlays for building height controls, flood constraints, and parking rates.

The paper makes three contributions. First, a methodological contribution: demonstrating that capacity-weighted, reliability-adjusted accessibility scoring materially changes which locations are identified as high-accessibility, and why that matters for planning decisions. Second, an

operationalisation contribution: linking PETAL bands directly to supportable bedroom density, providing a principled basis for height and density policy. Third, a novel spatial contribution: identifying where SEQ's highest-accessibility locations intersect with flood risk, with direct implications for infrastructure investment sequencing and climate adaptation planning.

The paper is structured as follows: Section 2 identifies the limitations of existing frequency-only accessibility metrics. Section 3 describes the PETAL methodology. Section 4 sets out the operationalisation framework linking accessibility scores to planning thresholds. Section 5 presents the application to SEQ. Section 6 compares PETAL with existing PTAL implementations, including Victoria and the original from Transport for London. Section 7 addresses transferability and application to other cities. Section 8 identifies limitations and future development. Section 9 provides a conclusion.

2. Limitations of frequency-only accessibility metrics

PTAL, as originally developed by Transport for London (TfL) and as currently implemented in Victoria and New South Wales, calculates accessibility based on service frequency and walking distance to stops. This approach captures important variation - a location served by four buses per hour is genuinely more accessible than one served by one bus per hour. But it treats all services of the same frequency as equivalent regardless of their people-moving capacity, reliability, or network position.

Four specific limitations are relevant to the Australian context:

2.1 Capacity equivalence assumption

Standard PTAL measures quantify accessibility based on service frequency without accounting for differences in vehicle capacity. Standard PTAL implicitly assumes that all services contribute equally to accessibility regardless of their passenger-carrying capability. In practice, however, substantial variation exists across modes. For example, a six-car heavy rail train in SEQ can accommodate approximately 450 seated passengers, compared to around 50 passengers on a standard bus. At equivalent headways, the train therefore provides an order of magnitude greater people-moving capacity.

Vehicle capacity limitations are particularly relevant in multi-modal networks where heavy rail, busway services, light rail, and conventional buses operate in parallel corridors. A frequency-based metric such as PTAL may therefore understate accessibility in areas served by high-capacity modes relative to those served by lower-capacity services at similar frequencies. These methodological issues reflect a broader limitation in accessibility measurement where service frequency alone is an incomplete proxy for transport supply as it does not capture capacity constraints (Tirachini et al., 2013; Currie, 2010).

2.2 Reliability conflation

Scheduled frequency overstates accessibility where services are frequently delayed or cancelled. A route scheduled at six services per hour but operating at 85% reliability effectively delivers 5.1 services per hour from a passenger perspective. Standard PTAL uses the scheduled frequency throughout. In SEQ, mode-specific on-time performance varies materially: G:link light rail operating on a dedicated corridor achieves approximately 98% reliability, while mixed-traffic bus routes average approximately 85%. Using the scheduled frequency for both overstates the relative accessibility of bus-dependent locations.

2.3 Route stacking inflation

Where multiple low-frequency routes serve the same stop, their frequencies aggregate in standard PTAL calculations. Ten routes each running once per hour score equivalently to a single route running ten times per hour, despite the passenger experience being substantially different. The former requires knowledge of individual timetables and offers no turn-up-and-go service; the latter enables spontaneous, unplanned travel. PTAL's accessibility index

applies a partial discount (0.5 weighting for secondary routes) but this does not adequately capture the diminishing utility of additional infrequent services.

2.4 Grid resolution and output format

Victoria's PTAL implementation uses a 200m grid anchored to the Melbourne GPO. New South Wales uses Australian Bureau of Statistics meshblocks, which are irregular, population-derived polygons that vary substantially in size between inner urban and outer suburban areas. Both produce static outputs: Victoria publishes a PDF map and a VicPlan GIS layer; NSW produces a similar static product. Neither implementation provides site-level transparency. A planner or developer cannot interrogate which stops are within the catchment, what services those stops receive, or how the score was derived for a specific location.

These limitations do not invalidate PTAL as a policy instrument. They identify where a more granular methodology would produce more defensible planning decisions, particularly in cities with pronounced modal hierarchies and high-stakes density policy settings.

3. PETAL methodology

PETAL adapts the TfL PTAL framework for SEQ's network structure and climate, with four core methodological enhancements: capacity weighting, reliability adjustment, route dominance decay, and bidirectional counting. The methodology uses open GTFS data from TransLink Queensland and a 100m x 100m grid covering Brisbane (15km radius from CBD) and the Gold Coast (LGA boundaries).

3.1 Capacity weighting by vehicle type

Each service is weighted by the passenger capacity of the vehicle type operating it, using manufacturer-published specifications. Table 1 sets out the capacity weights applied.

Table 1: PETAL capacity weights by vehicle type

Vehicle type	Capacity	Weight (relative to standard bus)
6-car train (NGR)	450	9.0x
G:link light rail	250	5.0x
Brisbane Metro (bi-articulated bus)	150	3.0x
Articulated bus (CityGlider)	75	1.5x
Standard bus	50	1.0x (baseline)
CityCat ferry	170	3.4x
KittyCat ferry	60	1.2x

G:link light rail receives a partial standing passenger credit reflecting manufacturer design intent and the smooth, dedicated corridor on which it operates. All other modes use seated capacity only. The rationale for capacity weighting is that a location served by ten trains per hour has fundamentally different development potential than one served by ten buses per hour, and a planning metric should reflect this.

3.2 Reliability adjustment

Scheduled frequency is discounted by mode-specific on-time performance factors derived from TransLink monthly performance reports (2023-2025 averages). Table 2 sets out the reliability factors applied.

Table 2: PETAL reliability factors by mode

Mode	Reliability factor	Basis
Heavy rail	0.95	~95% on time (TransLink performance data)
G:link light rail	0.98	~98% on time (dedicated corridor)
Busway	0.90	~90% on time (dedicated infrastructure, some traffic interaction)
Bus	0.85	~85% on time (mixed traffic)
Ferry	0.95	Similar to rail (dedicated infrastructure, no traffic interaction)

It is noted that reliability factors identified above are applied multiplicatively to scheduled frequency, consistent with the interpretation of accessibility as expected service availability. This formulation assumes that reliability affects the probability of service delivery rather than perceived waiting time, providing a tractable approximation for network-wide application.

3.3 Route dominance decay

Multiple routes serving the same location provide genuine redundancy but with diminishing returns. PETAL applies declining weights to additional routes beyond the primary service, preventing artificial score inflation where many infrequent services overlap. The decay schedule is 1st route 1.0x, 2nd 0.5x, 3rd 0.3x, 4th 0.2x, 5th 0.15x, 6th and beyond 0.1x. Route decay is applied separately within each direction of travel.

The decay function reflects diminishing marginal utility of additional services and is specified heuristically. While not empirically calibrated in this study, the structure is consistent with behavioural evidence on service frequency perception and transfer penalties. Future work will calibrate these coefficients using revealed preference data.

3.4 Bidirectional counting

Inbound and outbound services are counted separately. A location with 12 services per hour in each direction has genuinely superior accessibility to one with 12 services in a single direction only, as it enables return journeys. The best inbound and best outbound route each receive the 1.0x decay multiplier, with subsequent routes in each direction receiving declining weights.

3.5 Mode-specific catchments and distance estimation

Catchment distances reflect infrastructure permanence and observed passenger willingness to walk. Heavy rail, busway stations, and CityCat terminals use an 800m catchment; standard bus stops and KittyCat terminals use 400m. The 800m figure is consistent with international Transit Oriented Development (TOD) practice and Australian state government guidelines, reduced from TfL's 960m to reflect Brisbane's subtropical climate. Walking distances are estimated using Euclidean distance with a 1.3x network multiplier for standard pedestrian paths, and a 3.0x multiplier where the path crosses the Brisbane River (detected using Brisbane City Council's service catchment boundaries).

3.6 The PETAL formula

Total effective capacity for each grid cell is calculated as:

$$\text{Total Effective Capacity} = \Sigma(\text{Services/hr} \times \text{Capacity Weight} \times \text{Reliability Factor} \times \text{Decay Factor})$$

The calculation uses 7-9am weekday peak services from GTFS data. An example calculation for a cell served by Dutton Park station and Logan Road buses is shown in Table 3.

Table 3: Example PETAL calculation - Dutton Park station precinct

Route	Services per hour	Capacity weight	Reliability	Decay	PETAL Score	Standard PTAL Score
Train inbound	8	9.0x	0.95	1.00	68.4	8.0
Train outbound	8	9.0x	0.95	1.00	68.4	4.0
Bus 160 inbound	4	1.0x	0.85	0.50	2.0	2.0
Bus 160 outbound	4	1.0x	0.85	0.30	1.0	1.0
Total score					139.8	15.0
PETAL / Standard PTAL Band					PETAL 4A (Very Good)	Standard PTAL 2 (Moderate)

4. Operationalising accessibility scores as planning thresholds

A key differentiator of PETAL from standard PTAL implementations is the translation of band thresholds into bedroom density. This links the accessibility metric directly to planning decisions about building height and density, providing a principled basis for zoning policy rather than an arbitrary classification.

4.1 Bedroom density translation framework

The framework uses a catchment-based demand model. Each PETAL grid cell represents a 100m x 100m area. Transit demand is modelled across an 800m catchment (2.01 km² total area, 1.51 km² developable at 75% efficiency, allowing for roads, parks and other infrastructure). Based on established norms for residential apartment development, per 1,000m² developable site at 50% site coverage gives a 500m² floor plate, 80% efficiency to account for servicing and circulation yields 400m² usable area, accommodating four two-bedroom apartments at 100m² each, occupied by 2.5 persons per apartment (10 persons per storey).

Peak hour transit demand is calculated at 6% of the population (5.1% work commutes based on Queensland Household Travel Survey data, plus 0.5% education trips, rounded up for conservatism). Mode share assumptions reflect urban context: 20% in suburban locations, 40% in inner-ring mixed-use areas, and 50% in walkable transit-oriented locations where trip reduction effects dominate.

At a 50% redevelopment rate across the catchment, this yields 755 redeveloped sites. Peak hour transit demand per storey is calculated as 755 sites multiplied by 10 persons per storey, the 6% travel rate, and the applicable mode share. Table 4 sets out the resulting PETAL band thresholds and their correspondence to supportable building heights.

Table 4: PETAL bands, thresholds, and supportable heights

PETAL band	Description	Threshold (effective units/hr)	Supportable height	PT Mode share assumption
PETAL 4B	Excellent	≥240	30+ storeys (unlimited)	50%
PETAL 4A	Very Good	≥120	16-30 storeys	50%
PETAL 3	Good	≥50	8-15 storeys	50%
PETAL 2	Moderate	≥10	3-8 storeys	40%
PETAL 1	Poor	<10	Up to 3 storeys	20%

The PETAL 4A/4B split at 240 effective units per hour corresponds to the threshold at which turn-up-and-go service - requiring no timetable consultation - becomes reliable across all modes. Above this threshold, car ownership becomes genuinely optional for most trip purposes. Below it, timetable-dependent travel remains necessary for many residents. In the Brisbane study area, 75 cells (0.08%) reach PETAL 4B; 1,030 cells (1.07%) reach PETAL 4A.

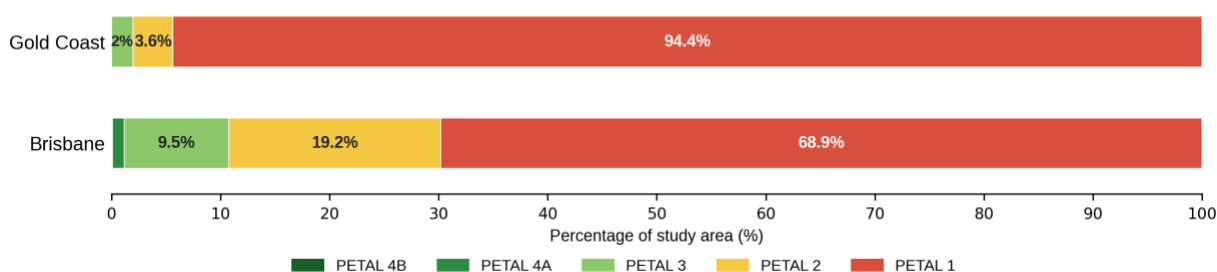
Note that the above mode share assumptions reflect observed variation in travel behaviour by urban context, supported by household travel survey data and broader evidence on transit-oriented development and accessibility-driven mode shift (Department of Transport and Main Roads, 2012; Department of Infrastructure, Transport, Regional Development and Communications, 2021).

5. Application to South East Queensland

5.1 Study area and data sources

PETAL Explorer covers Brisbane (15km radius from CBD) and the Gold Coast (City of Gold Coast LGA boundaries) using TransLink GTFS data (February 2026). The Brisbane study area comprises 96,119 grid cells after excluding waterways. The Gold Coast study area comprises 99,123 cells across the coastal LGA. PETAL Explorer operates at 100m grid resolution, making it a neighbourhood-scale analysis and visualisation tool. The overlay analysis in Sections 5.3 and 5.4 identifies systemic patterns across the network; site-specific planning, flood, and engineering assessments remain necessary for individual development decisions.

Figure 1: PETAL distribution by percentage of study area – Brisbane and Gold Coast.



5.2 Distribution of PETAL scores

Brisbane's PETAL distribution reflects its radial rail network and busway infrastructure. High-accessibility cells (PETAL 4A and 4B combined) represent just 1.2% of the study area (1,105 cells), concentrated along inner rail corridors and the busway network. Of these, only 75 cells reach PETAL 4B, reflecting the concentration of trunk transit infrastructure in a small number of inner-city locations.

The Gold Coast presents a contrasting pattern. 94.4% of the study area scores PETAL 1, with just 24 cells reaching PETAL 4A (0.02% of the study area) and none reaching PETAL 4B. The entire high-accessibility band is concentrated in the G:link light rail corridor, with suburban heavy rail station catchments scoring PETAL 3, and the remainder of the city, including major growth areas at Coomera and Pimpama, scores PETAL 1 or 2. The city's permissive zoning allows high-density development across much of the LGA, producing a systematic mismatch between where density is permitted and where transit can support it. Table 6 summarises the constraint analysis for the Gold Coast.

5.3 Land use mismatches

PETAL Explorer identifies two categories of land use mismatch: locations where transit accessibility is high but permitted density is low (transit-rich, planning-constrained), and locations where permitted density is high, but transit accessibility is poor (car-dependent density).

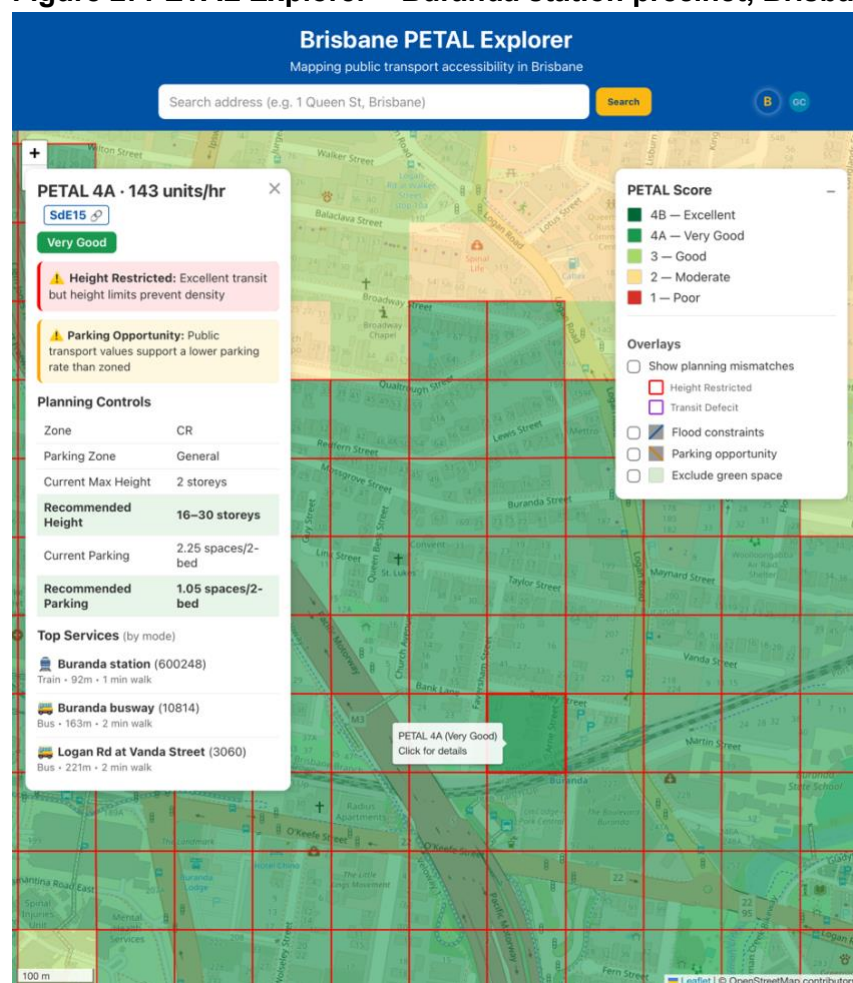
Of Brisbane's 1,105 PETAL 4A/4B cells, 473 are height-constrained - zoned below PETAL-recommended minimums. The mismatch is most acute at the highest accessibility tier: 51 of 75 PETAL 4B cells (68%) are under-zoned relative to their transit capacity, with 24 cells zoned at three storeys or below in locations where PETAL recommends 30-plus storeys. The inverse mismatch is also significant: 989 PETAL 1-2 cells are zoned for eight storeys or above, and 96 for fifteen storeys or above, in locations where transit capacity cannot support that density.

At historical uptake rates of approximately 1% per year of newly zoned capacity (Grattan Institute, 2025), misaligned zoning compounds over time in both directions: transit infrastructure capable of supporting 16 to 30-plus storeys stranded beneath two and three storey zoning, and car-dependent density generating infrastructure servicing costs estimated at up to 4 times those of equivalent infill development in established areas (SGS Economics and Planning, 2016). The costs accumulate regardless of which direction the misalignment runs.

The parking mismatch data reflects a broader misalignment between statutory minimums and transit-supported demand. 20,291 cells outside the City Core and Frame carry parking requirements that exceed PETAL-recommended rates, concentrated in PETAL 2-4 locations where transit access would support market-driven reductions. In locations requiring basement construction, each excess space adds an estimated \$80,000-\$100,000 to dwelling costs. A PETAL-informed approach would replace prescriptive minimums with accessibility-indexed rates, allowing the market to respond to actual transit capacity rather than uniform city-wide rules.

The Queensland Productivity Commission (2025, p. 230) found that 69 per cent of residential land within 1,000 metres of Brisbane train stations is zoned low density or protected by character overlays. PETAL Explorer makes this mismatch spatially legible at 100m resolution, enabling site-specific identification of where density policy diverges most sharply from network capacity.

Figure 2: PETAL Explorer – Buranda station precinct, Brisbane



Cell SdE15 scores PETAL 4A (143 effective units/hr). Character Residential (CR) zoning permits a maximum of 2 storeys where PETAL recommends 16-30 storeys. Source: petalexplorer.org (2026).

5.4 Flood constraint overlay

Table 5: Planning constraint analysis by PETAL band, Brisbane

PETAL band	Description	Total cells	Flood-constrained ¹	Height-constrained ²	Parking mismatch ³
PETAL 4B	Excellent	75	2	51	n/a
PETAL 4A	Very Good	1,030	38	422	345
PETAL 3	Good	9,133	504	n/a	6,379
PETAL 2	Moderate	18,438	605	n/a	13,567
PETAL 1	Poor	66,237	2,863	n/a	n/a
Total		96,119	4,932	473	20,291

¹ FPA1-3 represent flood events with 1-39% annual exceedance probability. FPA1 is the highest risk category (less than 1% annual chance); FPA3 the lowest mapped material constraint (10-39% annual chance).

² Current maximum height limit below PETAL-recommended minimum for that accessibility band.

³ Current minimum parking rate exceeding PETAL-recommended rate. PETAL 1 cells show no mismatch as existing rates align with low-accessibility assumptions. All PETAL 4B cells fall within the City Core and Frame parking zone where existing rates already reflect high accessibility.

Flood risk is distributed across all accessibility bands but is not a significant constraint on Brisbane's highest-accessibility locations: only 2 of 75 PETAL 4B cells and 38 of 1,030 PETAL 4A cells are flood-affected. Brisbane's highest-transit-capacity locations are almost entirely flood-free, strengthening the case for near-term densification where network capacity and land suitability align.

Areas with excellent transit but significant flood risk may require infrastructure investments - flood mitigation, elevated access, revised building typologies - before density targets are realistic. Where PETAL 4A/4B cells have minimal flood constraints, these should be prioritised for near-term densification over flood-affected sites requiring additional engineering. PETAL Explorer enables this prioritisation to be made transparently and at scale.

Table 6: Planning constraint analysis by PETAL band, Gold Coast

PETAL band	Description	Total cells	Flood-constrained ¹	Height-constrained ²	Parking mismatch ³
PETAL 4B	Excellent	0	n/a	n/a	n/a
PETAL 4A	Very Good	24	3	13	20
PETAL 3	Good	1,946	235	n/a	1,789
PETAL 2	Moderate	3,620	340	n/a	n/a
PETAL 1	Poor	93,533	13,881	n/a	n/a
Total		99,123	14,459	13	1,809

¹ Gold Coast Natural Hazards Overlay flood categories: Very High (9,416 cells), High (3,515 cells), Medium (1,528 cells). The Gold Coast's flood-constrained proportion (14.6%) is substantially higher than Brisbane's (5.1%), reflecting canal estate geography, low-lying coastal terrain, and Nerang River catchment patterns.

² Current maximum height limit below PETAL-recommended minimum for that band.

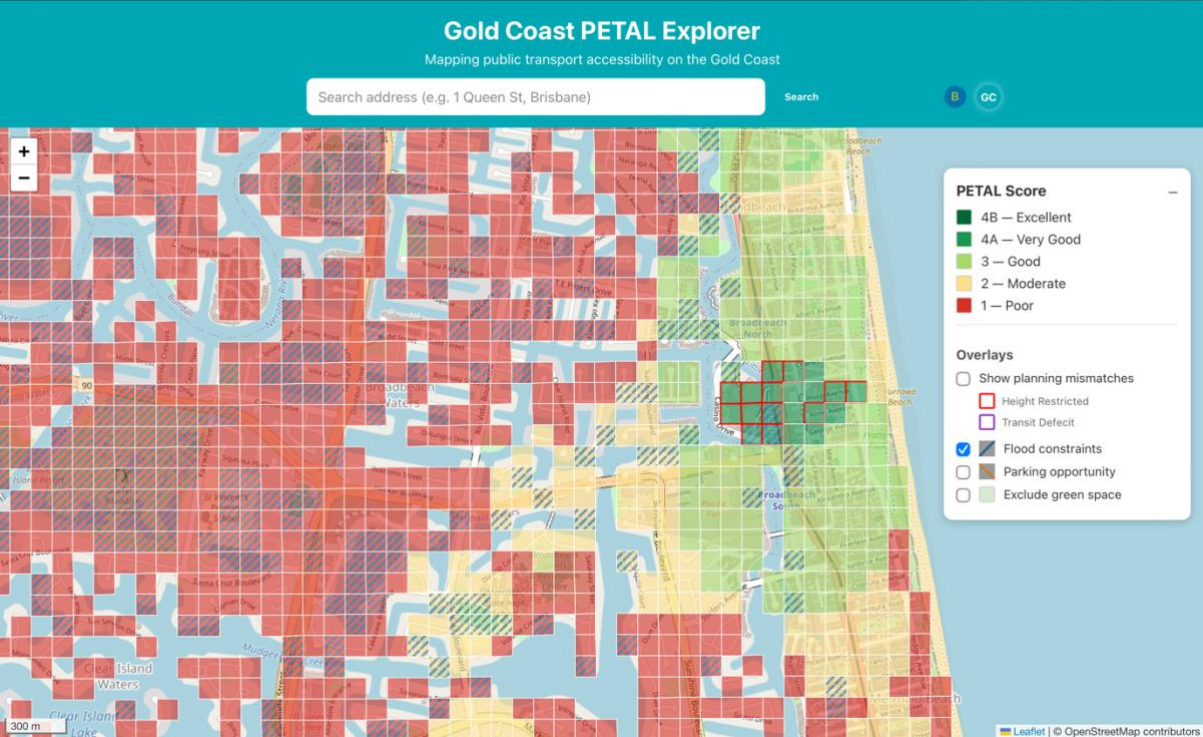
³ Gold Coast applies a uniform city-wide parking rate of 1.55 spaces per dwelling. Parking mismatch reflects locations where PETAL-recommended rates fall below this figure.

The Gold Coast's constraint profile differs from Brisbane's in two important respects. Flood risk is the dominant constraint: 14.6% of the study area is flood-affected compared to 5.1% in Brisbane, concentrated along the narrow coastal strip where development demand is highest and land supply most constrained. The most desirable locations to live are also the most flood-exposed, and the G:link light rail corridor that serves them represents the city's entire high-accessibility supply.

The broader pattern is density built ahead of infrastructure. Just 3% of Gold Coast trips use public transport, with 88% by private vehicle (ABC Gold Coast, 2025). Heavy rail station precincts scoring PETAL 3, identified as population growth areas, lack the surrounding density to support genuine mode shift: most residents face a choice between driving to limited commuter car parking, or boarding buses operating in mixed traffic subject to the same congestion.

The city has no rail connection to its airport, and the heavy rail connection to Brisbane operates from stations surrounded by low-density development at the southern extremity of the Queensland Rail network. The cancellation of G:link Stage 4, which would have extended light rail south from Burleigh Heads to the airport at Coolangatta, removed the one intervention that would have materially expanded the PETAL 4 cohort in the city's densest corridor. With no PETAL 4B cells and a single linear transit corridor, the Gold Coast's planning challenge is protecting and activating the limited accessible land that exists while transport and land use policy catch up with each other.

Figure 3: PETAL Explorer – Gold Coast, Broadbeach



G:link light rail corridor produces a concentrated band of PETAL 3-4A cells (green) along the coastal strip. Flood constraint overlay active (blue hatch). Source: petalexplorer.org (2026).

6. Comparison with existing PTAL implementations

6.1 Methodological comparison

Table 8 compares PETAL with Transport for London's original PTAL methodology and Victoria's December 2025 implementation.

Table 8: Methodological comparison of PTAL implementations

Feature	TfL PTAL (original)	Victoria (Dec 2025)	PETAL (this paper)
Capacity weighting	No	No	Yes - by vehicle type
Reliability adjustment	No	Bus +2 min, Train +0.75 min	Mode-specific factors (0.85-0.98)
Route decay	Partial (0.5 secondary)	Not specified	Graduated (1.0/0.5/0.3/0.2...)
Bidirectional counting	No	No	Yes
Rail catchment	960m	Not specified	800m (subtropical adjustment)
Grid resolution	Not applicable (stop-based)	200m	100m
Planning overlays	None	None (parking rates only)	Height, flood, parking
Output format	Static/GIS	PDF + VicPlan layer	Live searchable tool
Density operationalisation	No	No	Yes - bedroom density translation

Victoria's reliability adjustment uses absolute minute penalties rather than proportional factors: +2 minutes average wait time for buses, +0.75 minutes for trains. This approach captures the same underlying phenomenon - unreliable services require longer expected waits - but is not directly calibrated to observed on-time performance data. PETAL's proportional factors (0.85 for mixed-traffic bus, 0.98 for G:link) are derived from TransLink monthly performance reports and applied multiplicatively to scheduled frequency, producing a more direct link between measured performance and accessibility score.

The absence of capacity weighting in both TfL and Victorian implementations means that a bus corridor and a rail corridor at equivalent scheduled frequency score identically, despite the rail corridor typically having three to nine times the people-moving capacity. In SEQ's network, where the busway system uses articulated vehicles (1.5x standard bus capacity) and heavy rail uses six-car NGR sets (9.0x), this distinction is particularly material.

These differences are not incremental but structural: PETAL redefines accessibility from a service-count metric to a capacity-based supply metric.

6.2 Implementation and output format

Victoria's PTAL output is a static PDF map and a VicPlan GIS layer. The underlying scoring methodology is summarised in a three-page fact sheet but the model inputs -- which stops are within catchment, what services they receive, how the score was calculated for a specific address -- are not publicly accessible. A planner assessing a development application cannot verify or contest the score for a specific site.

PETAL Explorer addresses this directly. The tool is a live, searchable web application at petalexplorer.org. Clicking any 100m grid cell reveals the stops within the catchment, the services per hour on each route, the capacity weighting and reliability adjustment applied, and the resulting PETAL score. The score is fully auditable and contestable, which is important when accessibility claims are being used to support or oppose planning decisions.

Victoria uses a 200m grid; New South Wales uses ABS meshblocks, which vary from small inner-urban polygons to large outer-suburban areas. PETAL Explorer uses a consistent 100m grid across the study area. At 100m resolution, the tool captures accessibility differences within a single city block, relevant for permit assessment on a specific development site, not merely for neighbourhood-level planning analysis.

The planning overlay integration is the most significant practical difference. Neither Victoria nor New South Wales overlays building height controls, flood constraints, or parking rates onto their PTAL outputs. PETAL Explorer displays all three simultaneously with interactive toggles. The combination of the PETAL band, height mismatch, flood constraint, and parking excess on a single cell provides information directly applicable to a planning submission or infrastructure investment decision. This is the difference between an accessibility map and a planning decision support tool.

7. Transferability and application to other contexts

PETAL's methodological foundation, which includes open GTFS data, 100m grid generation, and capacity-weighted frequency scoring, is transferable to any city with publicly available transit schedule data. The GTFS standard is now adopted by transit agencies across Australia, New Zealand, North America, Europe, and much of Asia. The open-source codebase is available at github.com/brisbane-ptal/explorer.

7.1 Australian cities

Sydney and Melbourne are the natural first extensions. Both Transport for NSW and the Victorian Department of Transport and Planning have moved on PTAL in the past twelve

months, creating a direct policy context for comparison. The capacity weighting argument is more material in Australian networks than in London: Australian cities have starker modal hierarchies, with a wider gap between heavy rail capacity and standard bus capacity than exists in London's more uniform network.

Perth and Adelaide present strong candidates for extension given their rail-dominant network structures and active TOD reform agendas. Perth's rail network has seen substantial capacity investment in recent years; a PETAL analysis would reveal the degree to which zoning policy has kept pace with that investment. Adelaide's single planning system, replacing local government plan-making, creates a direct mechanism for state-level PTAL-based density controls.

Applying PETAL to Brisbane's sister cities also enables systematic comparison of SEQ's planning performance against other Australian jurisdictions, supporting the broader case for planning and land use policy reform.

7.2 International application

Any city with open GTFS data and a multi-modal transit network can implement PETAL. The analytical framework, linking capacity-weighted accessibility to supportable bedroom density via a peak hour demand model, is reproducible with local travel survey data on mode share and trip rates. The flood constraint overlay can be adapted to any mapped hazard layer.

The methodology is particularly relevant to cities in the Asia-Pacific region where rapid transit investment is outpacing planning reform: Singapore's MRT network, Auckland's City Rail Link, and Wellington's Let's Get Wellington Moving programme all create contexts where capacity-weighted accessibility measurement would produce materially different planning outputs from frequency-only approaches.

8. Limitations and future development

8.1 Known limitations

PETAL uses 7-9am weekday peak services only. Off-peak and weekend accessibility may differ substantially, particularly in outer suburban areas where peak-hour frequency is specifically enhanced. A location that scores PETAL 3 in the peak hour may score PETAL 1 on a Saturday evening.

The methodology uses scheduled capacity, not actual spare capacity. Services operating at crush load provide lower effective accessibility than the calculation suggests since additional passengers cannot board. This is particularly relevant for inner-Brisbane rail corridors during peak periods.

Walking distances are estimated using Euclidean distance with a network multiplier rather than actual pedestrian routing. The 1.3x multiplier is a simplification; actual network distances vary by location. The methodology does not account for gradient, surface quality, or pedestrian amenity - a 400m walk on a flat, shaded, well-maintained path is not equivalent to a 400m walk up a steep hill.

Active transport infrastructure: cycling networks, shared paths, and e-mobility options including e-bikes and e-scooters, are not captured in the current methodology. Last-mile connectivity via active transport can materially extend the effective catchment of transit stops. Incorporating active transport catchments represents a meaningful refinement to the mode-specific catchment assumptions and is identified as future work.

The flood overlay uses current Brisbane and Gold Coast flood mapping and does not incorporate climate change projections. Future flood risk under higher sea level and rainfall intensity scenarios may expand constrained areas significantly.

The route dominance decay coefficients and distance multipliers are informed estimates rather than empirically derived parameters. Comparison against revealed preference data, such as comparing PETAL scores with actual mode share and car ownership rates, would strengthen the framework.

8.2 Future development

Off-peak analysis producing separate PETAL scores for interpeak, evening, and weekend services would significantly extend the tool's policy relevance.

Destination-weighted accessibility: scoring not only frequency and capacity but also the employment and educational opportunities reachable within a given travel time would address a key limitation identified in the literature, namely the disconnect between service provision and actual travel demand. By incorporating the alignment between origins and destinations, such an approach would move PETAL beyond a supply-based measure toward a more meaningful representation of network utility. This would better capture user-relevant accessibility outcomes (Perera et al., 2025).

Brisbane's Cross River Rail stations are scheduled to open progressively from mid-2026. As the network's highest-capacity additions, these stations will materially shift PETAL scores across inner Brisbane catchments. A post-opening PETAL analysis would provide a timely measure of the extent to which zoning policy has kept pace with the \$19 billion infrastructure investment - and is planned for presentation alongside this paper at the November 2026 conference.

Extension to other Australian cities: beginning with Sydney and Melbourne given existing PTAL policy contexts, would enable direct comparison with state-level accessibility tools and strengthen the case for capacity-weighted measurement as a national standard. This represents a priority for future work.

Scenario testing: modelling accessibility impacts of proposed infrastructure such as Brisbane high frequency bus extensions and G:link expansions -- would enable PETAL to inform infrastructure investment decisions prospectively rather than describing existing conditions only.

9. Conclusions

PTAL adoption in Australian planning policy is a positive development. Victoria's December 2025 reform of clause 52.06, linking car parking requirements to measured transit accessibility, demonstrates that accessibility metrics can be successfully embedded in statutory planning instruments. This paper argues that the metric can be improved, and that improvement matters for the quality of planning decisions.

The central contribution of this paper is that accessibility is more accurately represented as capacity per hour rather than services per hour. A location served by ten trains per hour has nine times the people-moving capacity of one served by ten buses per hour. A methodology that treats them equivalently will systematically misidentify which locations can support density and which cannot. In cities with pronounced modal hierarchies, which describes most Australian cities, this error is not marginal.

PETAL addresses this weakness by incorporating capacity weighting, reliability adjustment, route dominance decay, and bidirectional counting into a single accessibility score operationalised against supportable bedroom density. Applied to South East Queensland, the methodology identifies land use mismatches that are both spatially specific and policy-actionable: locations where excellent transit capacity is being wasted by restrictive zoning, and locations where permissive zoning is generating density in car-dependent areas.

Bertaud (2018) argues that density restrictions near high-capacity transit corridors impose welfare costs by preventing the agglomeration benefits that transit investment is designed to

enable - a dynamic made spatially explicit by PETAL Explorer's mismatch analysis. The Queensland Productivity Commission (2025, p. 235) estimated that targeted zoning reforms in well-located areas around transport networks in South East Queensland could deliver net benefits of up to \$48 billion, more than double the \$18 billion projected from dispersed growth. The difference reflects infrastructure efficiency and transport cost savings: exactly the dimensions that PETAL is designed to measure. Making these mismatches legible and auditable at 100m resolution, with planning overlays for height controls, flood constraints, and parking rates, is the practical contribution of PETAL Explorer.

The methodology is open-source, built on universally available GTFS data, and transferable to any Australian or international city seeking a more rigorous basis for transit-informed land use policy. The framework, linking capacity-weighted accessibility to supportable density via a peak hour demand model, is reproducible with local travel survey data. Future work will extend the methodology to other Australian cities, refine decay coefficients and distance multipliers against revealed preference data, and examine off-peak accessibility and localised mode share calibration.

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Author contributions

Tristan Clark conceived the PETAL methodology, developed and implemented the PETAL explorer, conducted all spatial analysis, and was the primary author of this paper. Supun Perera contributed to the academic framing and literature review, provided peer review of the methodology, and anchored the work within transport research literature.

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